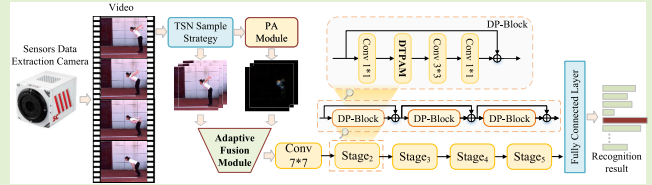


MIE-Net: Motion Information Enhancement Network for Fine-Grained Action Recognition Using RGB Sensors

Yutong Li^{ID}, Miao Ma^{ID}, Jie Wu^{ID}, Kaifang Yang, *Member, IEEE*, Zhao Pei, and Jie Ren^{ID}, *Member, IEEE*

Abstract—In recent years, action recognition has received widespread attention, which classifies actions by extracting features from kinds of sensor data. However, with the growing difficulty of identifying fine-grained actions, certain methods cannot learn sufficient motion and temporal information. Therefore, an effective information enhancement method is required to reason motion clues in video sequences. This article proposes an end-to-end video action recognition framework called the motion information enhancement network (MIE-Net), which consists of two innovative components. The first component, the adaptive fusion module (AFM), selectively extracts the relationships between original and motion-enhanced features to enhance the interaction among different feature information. The second component, a double pooling temporal attention module (DPTAM), implements temporal modeling to enhance subtle information during feature extraction. Finally, a standing long jump dataset (SLJD) containing over 1000 videos from 116 participants is collected by sensor camera, which differs from existing datasets in terms of strong background unbiasedness, to evaluate the effectiveness of our model robustly. Experimental results on SLJD, Something-Something v2, and Diving48 datasets demonstrate that the proposed MIE-Net outperforms most state-of-the-art methods. Our code is released at <https://github.com/li-stu-998/MIE-Net>.

Index Terms—Feature fusion, motion information enhancement, temporal modeling, video action recognition.



I. INTRODUCTION

ACTION recognition is a fundamental and challenging task in video understanding, playing an important role in other video tasks such as video captioning [1], temporal action detection [2], [3]. In recent years, scholars have proposed a series of advanced action recognition methods [4], [5], [6], [7], [8], [9], [10], [11] and achieved significant performance gains in recognizing different actions. However, there are still challenges in identifying similar actions with subtle action variations or differences. Focusing on this fine-grained

recognition problem, how to effectively combine local motion information to enhance global spatial-temporal information has become one of the critical solutions to this problem.

In early action recognition studies, stacked image sequences from red-green-blue (RGB) sensors are fed into convolutional neural network (CNN) models to extract the spatial information of action well. However, the temporal information presented by the image sequences is not effectively explored. Therefore, some studies [4], [5], [6] construct 3-D CNNs to learn spatiotemporal features, which expand the 2-D convolution to 3-D convolution. Nevertheless, 3-D CNNs suffer from high computational costs, lacking a reasonable strategy to address the calculation problem. Subsequently, Lin et al. [7], Liu et al. [8], Wang et al. [9], and Shao et al. [10] integrate attention units or modules on 2-D CNN to capture temporal information on the entire video. Lin et al. [7] proposed a shift operation that offsets part of the feature channels in the temporal dimension to better understand the spatial-temporal information contained in videos. Liu et al. [8] offered a two-level adaptive modeling scheme to capture diverse features. Although these methods effectively model spatial-temporal features, they lack motion information to guide the learning of local action variations.

Differing from the above-mentioned methods, some researchers suggest that utilizing short-term temporal features between adjacent frames, i.e., motion information, can

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help identify subtle actions. Simonyan and Zisserman [12] constructed a two-stream network to fuse spatial-oriented RGB features and temporal-oriented optical flow features. Wang et al. [13] proposed a generative framework to integrate complementary information from RGB and depth modalities. However, these methods still have certain limitations. The computational of optical flow is storage-demanding and time-consuming. The acquisition of depth images relies on other sensors, and those images are insensitive to illumination changes. Moreover, these methods are not implemented in an end-to-end manner. Hence, Zhang et al. [14] and Huang and Bors [15] explore lightweight motion features by RGB sensors and form an end-to-end training framework, reducing computation time and complementing motion information. However, they aggregate motion features and RGB features through simple fusion rules such as addition and concatenation, lacking interaction between multiple features. Also, they ignore the temporal modeling during feature extraction.

The above problems inspire us to investigate an end-to-end architecture that can effectively fuse motion information by considering the relationship between multiple features and further enhance the representation of spatial-temporal information by temporal modeling during feature extraction. Thus, we propose an end-to-end motion information enhancement network (MIE-Net) for fine-grained action recognition by incorporating RGB and motion information to capture subtle motion details and employing temporal attention along the channel dimension to improve the network's learning capabilities. In detail, we exploit the MIE features in PAN [14] to capture motion boundaries between video frames, then feed them and RGB features into the adaptive fusion module (AFM) to emphasize information interaction and information attention, which selectively extracts the relationships between RGB features and motion-enhanced features. Finally, the double pooling temporal attention module (DPTAM) enables the spatio-temporal convolution operator to perform temporal reasoning along the channel dimension. To evaluate our approach, we compare it with other state-of-the-art action recognition methods on our dataset standing long jump dataset (SLJD), Something-Something v2 [16], and Diving48 [17]. The experimental results demonstrate superior performance in recognition accuracy and efficiency.

The contributions of this article are fourfold.

- 1) We propose a MIE-Net for fine-grained action recognition, which aims to incorporate RGB and motion features into a unified framework. Our model is able to learn more motion and spatial-temporal information from RGB and motion features in an end-to-end fashion.
- 2) We design an AFM to guide the fusion of RGB and motion features by considering the relationship between the features. Our fusion module emphasizes the interactions between RGB features and motion-enhanced features to adaptively enhance the spatio-temporal representation of the fused features.
- 3) We propose a DPTAM to extract subtle features by modeling the representation of spatial features in channel and temporal dimensions, which can be flexibly inserted

into the 2-D CNN backbone without overly changing the topology of the network.

- 4) Some standing long jump (SLJ) videos are collected by the sensor camera to further evaluate our model's effectiveness and application scenarios. We evaluate the MIE-Net on Something-Something v2, Diving48, and our created SLJD, and multiple datasets show impressive results.

The manuscript is organized as follows. Section II introduces related work on state-of-the-art action recognition methodologies. Section III explains the details of our proposed framework. Section IV describes the construction process of the proposed dataset. The experiments and the implementation details are illustrated in Section V. Section VI is a conclusion of this article.

II. RELATED WORK

A. Action Recognition Based on Temporal Modeling

The early method for exploiting temporal information in action recognition was C3D [4], which learns spatio-temporal features effectively by introducing the temporal dimension to 2-D kernels. I3D [5] expands 2-D kernels to 3-D kernels in the inception module and copies 2-D network pre-trained weights to 3-D network. However, the computation of 3-D convolutional kernels has an expensive cost. Some variants of 3-D CNN like P3D [6], R(2+1)D [18], S3D [19], and CSN [20] have been subsequently proposed, which decompose 3-D convolution into a temporal 1-D convolution and a spatial 2-D convolution to connect time and space. These variant structures or methods reduce computational complexity and improve accuracy to some extent. Unlike the above methods, Feichtenhofer [22] expanded a tiny 2-D image classification architecture across several axes in space, time, channels, and depth to optimize the execution efficiency of the network. However, 3-D CNN and a series of variants still have a significant computational burden.

Subsequently, some studies have attempted to construct attention units with fast inference abilities on 2-D CNN to model temporal and spatial information. TSM [7] achieves temporal modeling by aggregating the temporal features with the parameter-free temporal shift operation. Moreover, several works employ spatial, temporal, channel, and motion excitation to spatio-temporal information, such as TAM [8], Action-Net [9], and GCT [22]. Wang et al. [23] pioneered a self-attention NonLocal (NL) block for visual recognition tasks to model pixel-level spatial-temporal dependency. Shen et al. [24] implemented spatiotemporal encoding with dynamic enhancement. Hao et al. [25] refined the video spatial-temporal features by inserting one attention unit into the other attention module.

B. Action Recognition Based on Motion Representation

Some researchers argue that the motion information presented by adjacent frames can be used as complementary features to guide and identify subtle action variations. Optical flow and depth maps, as a representation of short-term motion information, have been successfully applied to action

recognition tasks. For example, two-stream convolutional networks [12] and TSN [26] process the spatial stream of RGB frames and the temporal stream of optical flow, respectively, and the final prediction of human action is obtained by fusing the results of each stream. These methods suffer from problems that optical flow extraction is time-consuming and storage-demanding, making it inefficient for large-scale video datasets.

Subsequently, some methods explore end-to-end trainable motion information by utilizing feature information between adjacent frames. For example, OFF [27] and EMV [28] use the modified optical flow units to guide motion feature learning. PAN [14], FineCoarse [15], and EAN [29] form a trainable end-to-end framework by introducing new features. These methods supplement neglected motion information and avoid high storage requirements caused by optical flow. However, these methods aggregate new features and RGB features by simple rules, such as addition and concatenation fusion, without considering the characteristics of each feature. They lack the interaction between features and cannot effectively combine the complementary information of RGB and other features.

C. Multimodal Fusion

In addition to utilizing data from RGB sensors, some researchers perform vision tasks based on data from different modalities. According to the source of sensor, multimodal data can be divided into optical flow, depth, skeleton, inertial, etc. Since each modal data generally contains specific information, it is a natural choice to utilize the complementary advantages of different data modalities by constructing effective fusion methods. In order to fully fuse features, some methods [30], [31], [32], [33], [34], [35], [36] utilized attention mechanisms to integrate multi-modal features and promote the interaction between features. For example, Wang et al. [30] proposed a multi-stream interaction network, simultaneously exploring the dynamics of human skeletons, objects, and the interaction between humans and objects. Zhou et al. [31] introduced a cross-modal fusion mechanism to capture cross-modal spatiotemporal information from RGB and RGB-D data. Su et al. [32] proposed a multi-modal split attention fusion method to emphasize more contributive features across all modalities.

In summary, existing action recognition methods do not fully exploit the potential of motion representation and temporal modeling to improve the efficiency of fine-grained action recognition. Therefore, this article proposes a method to enhance fine-grained action recognition using motion information, which aims to effectively utilize motion features, promote their fusion with RGB features, and further enhance the global spatial-temporal information representation of the fused features.

III. METHOD

A. Framework Overview

Denote a dataset as $\text{Dataset} = \{D_{\text{train}}, D_{\text{test}}\}$, the corresponding annotation can be depicted as $Y = \{y_{a_s}\}_{a_s=1}^{A_s}$,

A_s indicates action class, $a_s = 1, \dots, A_s$. Our goal is to train a model to predict the class scores of each video by Top-1 and Top-5 accuracy. The notations used in the section include N (batch size), T (number of segments), C (channels), H (height), and W (width).

In this work, we propose MIE-Net as an end-to-end video action recognition framework, there are five steps as illustrated in Fig. 1. The first step divides the input video sequence V into T segments with equal length $V_{\text{each}} = \{S_t\}_{t=1}^T$ by following the sampling strategy of TSN [26], and consecutive four frames are randomly selected from each segment for stacking in order to get the set of adjacent frames $\{I_t\}_{t=1}^T$. In the second step, we select the first frame of each “four frames” to generate the RGB features $X_{\text{RGB}} = \{I_{t[1]}\}_{t=1}^T$. Then, persistent appearance (PA) module in PAN [14] processes each “four frames” $\{I_{t[k]}\}_{k=1}^4$ in $\{I_t\}_{t=1}^T$ to capture saliency-enhanced map of low-level features and obtain the motion-enhanced features $X_{\text{MIE}} = \{MIE_t\}_{t=1}^T$, with the aim of perceiving small displacements between adjacent frames and highlighting subtle motion variations at boundaries. In the third step, AFM is used to guide the fusion of X_{RGB} and X_{MIE} for emphasizing their interaction. In the fourth step, we feed the fused features into the backbone, and we achieve temporal modeling by embedding DPTAM in the ResNet-50 [37] to enhance the feature extraction ability of the network. Finally, the class scores are output by a fully connected layer. Here, the input tensor of AFM is all 4-D ($N \times T, C, H, W$). We reshape a 4-D tensor into a 5-D tensor (N, T, C, H, W) to note the relationships between temporal, channel, and spatial dimensions before performing DPTAM. Once the DPTAM is completed, the 5-D output tensor is reshaped to 4-D.

B. PA Module

To improve the accuracy of recognition, it is important to mine subtle action information that is prone to be missed. We extract motion information by using the PA module in PAN [14]. The main objective of the PA module is to extract small displacements at the motion boundaries of actions from the sequence of frames. It captures the motion information presented between adjacent RGB frames by calculating the pixel level differences between low-level feature maps, as shown in Fig. 2. In contrast to optical flow, this motion representation only captures the motion variation at a certain point in feature space without considering the direction of the movement, making its calculation faster and more effective. For any set of input consecutive frames $\{I_{t[k]}\}_{k=1}^4$ in $\{I_t\}_{t=1}^T$, the main process is as follows.

Given consecutive four frames $\{I_{t[k]}\}_{k=1}^4 \in R^{H_f \times W_f \times 3}$ with H_f , W_f and 3 being their height, width and channel number of each frame. First, low-level feature maps $\{F_{t[k]}\}_{k=1}^4$ are generated through 7×7 convolution layer without reducing spatial resolution, which can be represented as

$$F_{t[k]} = \text{Conv}(I_{t[k]}) \quad (1)$$

where $F_{t[k]} \in R^{H_f \times W_f \times C_f}$, C_f is the output channel number of 8, $\text{Conv}(\cdot)$ denotes the convolution operation with stride=1 and padding=3. Then, each adjacent low-level feature map performs pixel-level euclidean difference to obtain

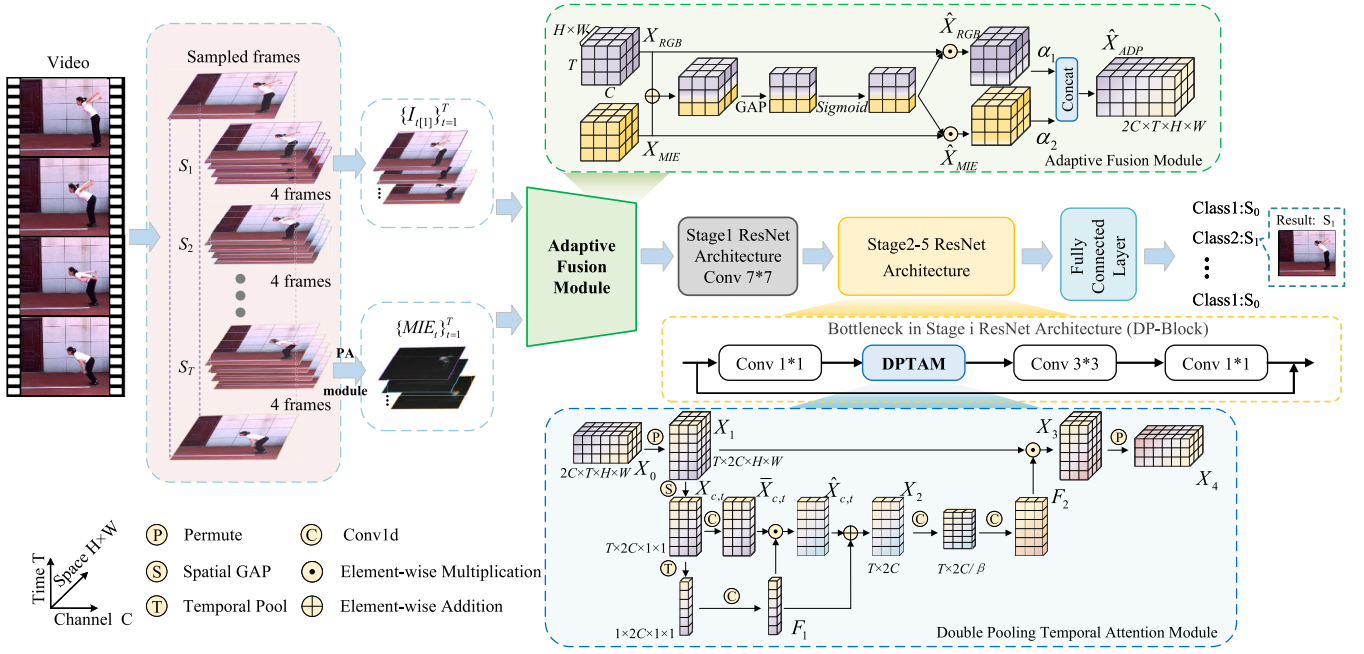


Fig. 1. Illustration of the proposed MIE-Net. Given an input video V , we perform video frame sampling and create the set of RGB and motion-enhanced features. Next, we emphasize the interaction between two features via AFM. Then, we feed the fused features into the backbone and implement temporal modeling by DPTAM during feature extraction. Finally, the class scores are output by a fully connected layer.

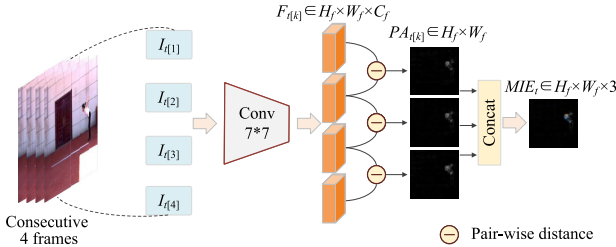


Fig. 2. Illustrations of PA module.

$\{PA_{t[k]}\}_{k=1}^3 \in R^{H_f \times W_f}$. Formulated as

$$PA_{t[k]} = \sqrt{\sum_{c_f=1}^{C_f} (F_{t[k+1]}(h_f, w_f, c_f) - F_{t[k]}(h_f, w_f, c_f))^2} \quad (2)$$

where $F_{t[k]}(h_f, w_f, c_f)$ denotes the pixel value of low-level feature map $F_{t[k]}$. Finally, each $PA_{t[k]}$ in $\{PA_{t[k]}\}_{k=1}^3 \in R^{H_f \times W_f}$ is concatenated along the channel dimension to form the motion clues $MIE_t \in R^{H_f \times W_f \times 3}$

$$MIE_t = \Phi_{k=1}^3 (PA_{t[k]}) \quad (3)$$

where $\Phi_{k=1}^3(\cdot)$ refers to the cumulative channel concatenation function that chronologically concatenate the input tensor along the channel dimension.

By processing the set of adjacent frames $\{I_t\}_{t=1}^T$ through the PA module, we obtain motion-enhanced features $X_{MIE} = \{MIE_t\}_{t=1}^T$ with the same dimensions as the RGB features $X_{RGB} = \{I_{rgb}\}_{t=1}^T$, which are used as additional input features to be fed into AFM along with the original features.

C. Adaptive Fusion Module

Currently, the fundamental methods of feature fusion can be divided into element-wise addition and concatenation operation, widely used in vision tasks. Suppose that two feature vectors to be fused are $x_1 \in R^m$, $x_2 \in R^n$, feature transformation vector is $W \in R^{m \times n}$, and the fused feature is X . The element-wise addition is defined as

$$\begin{cases} X = x_1 + x_2, & \text{if } m = n \\ X = x_1 + Wx_2, & \text{else} \end{cases}, \quad X \in R^m \quad (4)$$

where input features are required to be of the same dimension. If the dimensions differ, a linear transformation normalization is required. In addition, the concatenation operation is formulated as

$$X = \text{Concat}(x_1, x_2), \quad X \in R^{m+n} \quad (5)$$

where $\text{Concat}(\cdot, \cdot)$ indicates concatenation operation along the channel dimension, whether the dimension of input features is the same or not. However, both of them ignore the inherent relationships and interactions between features.

Unlike simple fusion methods above, SKNets [38] adjusts kernel sizes selectively to process information, and CBAM [39] imposes channel and spatial attention to self-recalibrate feature maps. Remarkably, inspired by the idea of SKNets [38] and CBAM [39], we design the AFM to adaptively fuse multiple features as illustrated in Fig. 1. Specifically, we implement the fusion process via three steps. First, we integrate the information from RGB and MIE features by element-wise addition. Then, we generate channel-wise attention weights by compressing the spatial dimension of the fused features to guide the original RGB and MIE features and enhance the spatial global information of modified features.

We fuse the features from multiple branches via an element-level addition. Formulated as

where X_{RGB} and X_{MIE} represent RGB features and motion-enhanced features, respectively, $\oplus$ denotes element-level addition. Then, we yield channel-wise attention weight $M \in R^C$ simply by global average pooling with a sigmoid activation, which achieves the compression of spatial information. Formulated as

where h, w is the index of different dimensions (in height and width), $Sigmoid(\cdot)$ denotes activation function.

The channel-wise attention weight M is multiplied by the original RGB features and the motion-enhanced features respectively to create modified features $\hat{X}_{\text{RGB}}$ and $\hat{X}_{\text{MIE}}$ that maintain the same dimensions as the original features. Formulated as

$$\hat{X}_{\text{MIE}} = M \odot X_{\text{MIE}} \quad (9)$$

where $\odot$ denotes element-level multiplication.

In order to better employ the modified features, we concatenate them with different learning weights. The calculation is expressed as

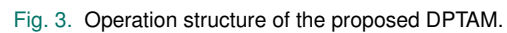
$$\alpha_i = \frac{e^{\omega_i}}{\sum_j e^{\omega_j}} \quad (11)$$

where α_i is a pair of normalized weights, $\sum_i \alpha_i = 1$, ω_i is an initial weight vector, ω_j is a feature weight vector, $i = 1, 2$, $j = 1, 2$.

D. Double Pooling Temporal Attention Module

Currently, attention mechanism in deep learning is extensively used in various tasks such as natural language processing (NLP) [40], computer vision (CV) [41], and speech recognition (SR) [42]. As for CV, its basic idea is to enable the model to ignore irrelevant regions of information and extract more important information without more computation or storage required.

Inspired by the attention mechanism, SE-Net [43] improves the modeling ability of the network by re-weighting channel-level features. Not the same as image processing, both spatial features and temporal features are considered at the same time in video understanding. Therefore, we design the DPTAM to model the interdependencies among time and channels, which can be plug-and-play in the backbone network to refine the feature information. The design of the DPTAM is shown in Figs. 1 and 3.



Given an input $X_0 \in R^{N \times T \times C \times H \times W}$, tensor X_0 is permuted to $X_1 \in R^{N \times C \times T \times H \times W}$. Formulated as

$$X_1 = \text{Permute}(X_0) \quad (12)$$

where $\text{Permute}(\cdot)$ denotes tensor permutation operation. Then, we aggregate X_1 spatial information into a global descriptor $X_{c,t}$, which is achieved by a squeeze operation to extract the temporal-channel information in spatial dimensions. The calculation is expressed as

$$X_{c,t} = \frac{1}{H \times W} \sum_u^H \sum_v^W X_1[:, :, :, u, v] \quad (13)$$

where u, v is the index of different dimensions (in height and width), $X_{c,t} \in R^{N \times C \times T \times 1 \times 1}$. Next, we generate $\tilde{X}_{c,t}$ by a 1-D convolution to enhance channel dependencies in the temporal dimension. Formulated as

$$\bar{X}_{c,t} = \text{Conv1d}(X_{c,t}, 1, C) \quad (14)$$

where $\text{Conv1d}(\cdot, 1, C)$ is 1-D convolution with the kernel size of 1, $\bar{X}_{c,t} \in R^{N \times C \times T \times 1 \times 1}$.

The global channel attention weight F_1 is calculated by a temporal pooling and a 1-D convolution on the global descriptor $X_{c,t}$, in order to excite the channel-wise information associated with temporal dimensions. Formulated as

$$F_1 = \text{Conv1d}\left(\frac{1}{T} \sum_{t=1}^T X_{c,t}[:, :, t], 1, C\right) \quad (15)$$

where t is the index of time dimensions. We multiply the feature $\bar{X}_{c,t}$ with attention weight F_1 to obtain $\hat{X}_{c,t}$

$$\hat{X}_{c,t} = \bar{X}_{c,t} \odot F_1 \quad (16)$$

where $\odot$ denotes element-level multiplication, and we add $\hat{X}_{c,t}$ with attention weight F_1 to generate X_2

$$X_2 = \hat{X}_{c,t} \oplus F_1 \quad (17)$$

where $\oplus$ denotes element-level addition, $X_2 \in R^{N \times C \times T \times 1 \times 1}$. Our purpose is to enhance the channel information of $\hat{X}_{c,t}$ in complement to the local 1-D convolution and further incorporate the global channel-wise information so that X_2 obtains a larger inter-frame field of view while maintaining the same receptive field as the global descriptor.

We generate the importance weight F_2 which is sensitive to temporal position. Formulated as

$$F_2 = \text{Sigmoid} \left(\text{Conv1d} \left(\delta \left(\text{Conv1d} \left(X_2, K, \frac{C}{\beta} \right) \right), 1, C \right) \right) \quad (18)$$

where δ denotes activation function ReLU, K is the kernel size of 3, β is a hyper-parameter of 4. Then, importance weight F_2 is multiplied with X_1 to obtain X_3

$$X_3 = X_1 \odot F_2 \quad (19)$$

where $X_3 \in R^{N \times C \times T \times H \times W}$. The short-term frame information is learned from a 1-D convolution with the local temporal field. Thus, all channels with integrated spatial and temporal information are linearly weighted to emphasize channel interaction.

In order to maintain the same size with the original input, before features are fed to the next block, X_3 is permuted to $X_4 \in R^{N \times T \times C \times H \times W}$. Formulated as

$$X_4 = \text{Permute}(X_3). \quad (20)$$

The DPTAM is embedded after the first 1×1 convolution of the Bottleneck in ResNet-50 [37] backbone (Stage2-Stage5). The module employs spatial operations to adjust the global information and temporal operations to capture the local receptive field, resulting in the effective calibration of spatio-temporal information. It could be flexibly inserted into existing 2-D CNN to improve the network's ability without overly changing the topology of ResNet [37].

Thus, we propose two ideal auxiliary modules in the MIE-Net. AFM is expected to flexibly fuse multi-feature information and accurately extract regions of interest, which solves the problem of insufficient feature interaction in video classification. DPTAM is expected to enhance the integrated motion information on the channel and temporal dimensions, further improving video classification accuracy.

IV. STANDING LONG JUMP DATASET

SLJ is an important measure of human physical abilities, which is considered an essential component of physical fitness tests worldwide. We collect some real-world SLJ videos by the sensor camera in order to validate the application scenarios of the model.

A. Data Sources

Considering the fast and transient characteristics of the SLJ, we use a high-speed camera to acquire the process of SLJ and have 116 volunteers participate in this data acquisition. The videos include three different backgrounds with two movement directions, jumping from left to right or jumping from right to left. We collect 154 SLJ raw videos at first and then remove the videos that are difficult to recognize. Finally, we get 107 untrimmed videos. The duration of each video is 13 s, with a video resolution of 1280×1024 at 15 frames/s.

B. Expert Definition

We invite sports experts from the school of Physical Education to decompose the process of SLJ, and then discover that actions always include arm swing, takeoff, flight, and landing. At last, we select nine moments that have a direct influence on jump distance. The nine actions corresponding to these moments are sequentially named as arm at backward highest, heel off ground, feet off ground, arm backward stretch, thigh vertical to the ground, knee at the highest, heel landing, feet landing, and hip at the lowest. Fig. 4 presents the details of each action. We denote the nine moments as $T_1, T_2, T_3, T_4, T_5, T_6, T_7, T_8$, and T_9 . According to the nine moments, we decompose the SLJ into ten stages, which are labeled as $S_0, S_1, S_2, S_3, S_4, S_5, S_6, S_7, S_8$, and S_9 . The stage before T_1 is denoted as the preparation stage S_0 , while the clip after T_9 is denoted as buffering stage S_9 . Finally, we separate an SLJ process into ten stages with nine key actions.

C. Data Labeling

According to ten stages defined in Section IV (B), annotators label nine key moments $T_1, T_2, T_3, T_4, T_5, T_6, T_7, T_8$, and T_9 for each video. To ensure correct annotations, we ask the sports experts to check all the labels and adjust the wrong label. After that, we utilize OpenCV to split SLJ video sequences into ten classes according to the labeled moments.

Finally, based on the previous definition of SLJ, we split each video into ten stages. Thus we can get 1070 video clips from 107 videos, and each video clip represents one of the ten stages of SLJ. Table I describes the details of each stage. We randomly split the dataset into 780 training videos and 290 testing ones. To sum up, the video characteristics of the SLJ include: 1) the movement is transient, no more than 5 s; 2) every jumping always sequentially covers ten stages (classes), where the last frame of one class is associated with the first frame of the next class for S_0 – S_9 ; 3) the duration of different classes varies from person to person, affecting the performance of the SLJ; and 4) subtle action differences are mainly embodied in the spatial semantic information of humans' limbs.

V. EXPERIMENTS

A. Datasets

1) *Standing Long Jump Dataset*: As mentioned in Section IV, SLJD is a classification dataset on the SLJ, containing over one thousand video clips in ten classes. Each untrimmed video includes six main procedures for

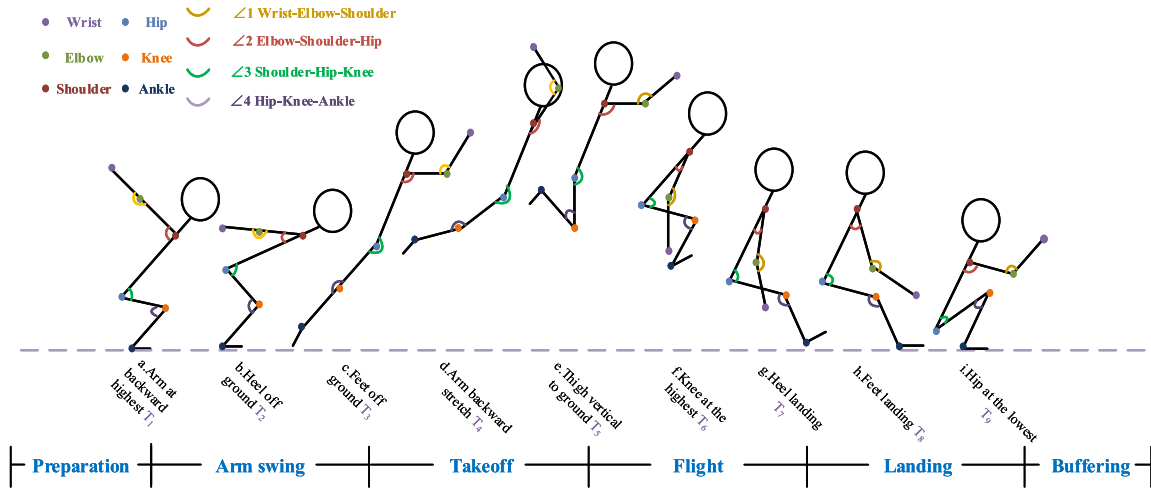


Fig. 4. Sports experts consider that nine key moments have a direct influence on the final jumping distance. The nine actions corresponding to these moments are sequentially named as: arm at backward highest, heel off ground, feet off ground, arm backward stretch, thigh vertical to ground, knee at the highest, heel landing, feet landing, and hip at the lowest.

TABLE I
AVERAGE NUMBER OF FRAMES AND AVERAGE
DURATION OF EACH STAGE

Stage	Frames ¹	Duration(s) ²
S ₀ : Preparation	54	1.80
S ₁ : Arm at backward highest → Heel off ground	19.5	0.65
S ₂ : Heel off ground → Feet off ground	21.3	0.71
S ₃ : Feet off ground → Arm backward stretch	3.5	0.11
S ₄ : Arm backward stretch → Thigh vertical to ground	7.3	0.24
S ₅ : Thigh vertical to ground → Knee at the highest	12.1	0.40
S ₆ : Knee at the highest → Heel landing	8.3	0.27
S ₇ : Heel landing → Feet landing	3.5	0.11
S ₈ : Feet landing → Hip at the lowest	13.7	0.46
S ₉ : Buffering	55.8	1.86

preparation, arm swing, takeoff, flight, landing, and buffering. Compared with the spatial semantic information in the scene, the subtle dynamic information between the limbs should be focused, which is conducive to the accurate analysis of SLJ.

2) *Something-Something v2 (SSv2)*: SSv2 [16] is a large-scale video dataset that contains over 220 000 video clips depicting human interaction with everyday objects. The dataset has 174 different action classes, and each video clip is associated with one of these classes. Its training set contains about 160 000 video clips, the validation set includes about 20 000 video clips, and the test set contains about 40 000 video clips. Each video clip in the dataset lasts around 2–6 s. The dataset emphasizes a model's ability to recognize fine-grained human actions due to the variations in objects and backgrounds.

3) *Diving48*: Diving48 [17] is a fine-grained video dataset of competitive diving containing 18 thousand video clips for 48 unambiguous dive sequences defined by FINA. All 48 sequences contain three stages, takeoff, flight, and entry.

Most diving scenes are similar, which makes it challenging for action recognition tasks. The actions of the Diving48 dataset have low bias, thus both long-time temporal dynamics and subtle information should be utilized, which differs from the Kinetics [44] dataset. Since the original Diving48 [17] dataset has some problems in dive annotations, we use the annotation file updated in 2020, where poorly segmented videos were removed by the official website.

B. Experimental Setting

We evaluate our approach on NVIDIA GeForce GTX 3 × 1080Ti or 1 × 3090 GPUs, and the prototype system is implemented by using the PyTorch framework. We leverage the ResNet-50 [37] as the backbone, which was pre-trained on ImageNet [45]. For all experiments, SGD with a momentum of 0.9 is used as the optimizer, and dropout is set to 0.6 using a batch size of 8. All sampled adjacent frames are processed into the PA module to obtain motion-enhanced features. Simultaneously, the first frame of every “four frames” is used as initial RGB features. We apply center cropping and scale-jittering for each sampled frame, resize the short side length to 256 pixels, and then use random cropping to resize the frame to 224 × 224 pixels.

1) *Model Training*: Our MIE-Net model is trained on the SLJD dataset for 60 epochs with an initial learning rate of 0.0025, decayed by a factor of 10 at 36 and 50 epochs, respectively. Meanwhile, we train our MIE-Net model on SSv2 dataset for 60 epochs with an initial learning rate of 0.00125 and the learning rate drops by ten at epochs 36 and 50. On the Diving48 dataset, we use an initial learning rate of 0.00125 for 30 epochs and decay it by a factor of 10 at 18 and 25 epochs. The weight decay of the optimizer is set to 1e−4 for each dataset.

2) *Model Testing*: TSM [7] and TAM [8] inspire us to take the protocol (one clip × center crop) for testing. Hence, we crop the center 224 × 224 pixels from the original frames resized to 256 × 256 for the three datasets.

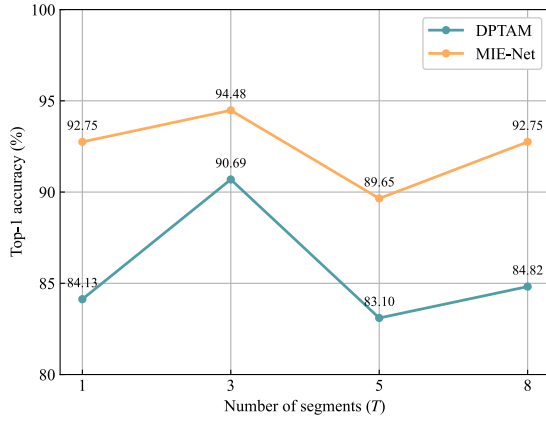


Fig. 5. Evaluation of the number of segments (T) in sampling strategy.

TABLE II
EVALUATION OF THE POSITION OF THE DPTAM

Position	Params	SLJD Top-1(%)	Diving48 Top-1(%)
DPTAM-1	26.04M	90.69	77.61
DPTAM-2	26.04M	88.96	76.80
DPTAM-3	63.77M	86.20	78.58

C. Ablation Studies

To investigate the effectiveness of the proposed AFM and DPTAM in MIE-Net, we report ablation experiments on SLJD dataset and Diving48 [17] dataset, respectively.

1) *Number of Segments (T) in Sampling Strategy*: The video characteristics of SLJ and the average duration of each stage are described in Section IV. The average number of frames and average duration of each stage on the SLJD dataset are listed in Table I. It is easy to see that the average frame number of all stages is 19.9. Experiments on the number of sampled frames can be referenced in FineGym [46], we first explore the segment numbers by the DPTAM inserted in ResNet-50 [37] on the SLJD dataset. As shown in Fig. 5, we have conclusive results in which $T = 3$ is better. Experimenting on MIE-Net, $T = 3$ is also preferred. Therefore, we apply it in all following experiments. Diving48 [17] has been widely used as a popular sports dataset, so we set the number of segments to eight by following the sampling strategy of TSM [7].

2) *Position of the DPTAM*: We insert the DPTAM into the residual block to enhance the temporal modeling capability of the ResNet [37]. The typical residual block of the ResNet-50 consists of 1×1 convolution, 3×3 convolution, and 1×1 convolution. We evaluate the performance of DPTAM in different positions. When DPTAM is inserted after the first 1×1 convolution, 3×3 convolution, and the last 1×1 convolution, it is referred to as DPTAM-1, DPTAM-2, and DPTAM-3, respectively. As shown in Table II, the performance of DPTAM-1 on SLJD dataset performs best, while DPTAM-3 is superior to DPTAM-1 on Diving48 [17]. The reason is that more channels (restored by the last 1×1 convolution) are utilized in the temporal dimension. At the same time, DPTAM-3 causes three times parameters compared with other inserted positions. By making a trade-off between parameter and accuracy, DPTAM-1 is selected as the inserted

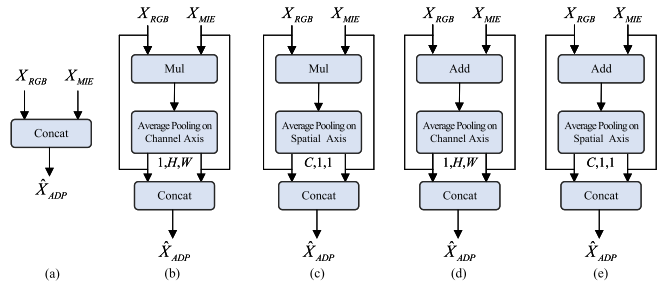


Fig. 6. Five implementations of feature connection.

TABLE III
EVALUATION OF THE DIFFERENT FUSION STRATEGIES

Method	SLJD Top-1(%)	Diving48 Top-1(%)
MIE&RGB_CAT	92.06	75.58
MIE&RGB_MC_CAT	93.79	78.52
MIE&RGB_MS_CAT	92.41	79.31
MIE&RGB_AC_CAT	93.10	78.57
MIE&RGB_AS_CAT	94.48	80.20

TABLE IV
COMPARISON WITH SE BLOCK AND NONLOCAL BLOCK

Model			Params	SLJD	Diving48
Baseline	AFM	Block		Top-1(%)	Top-1(%)
ResNet-50	×	×	23.52M	65.17	73.75
ResNet-50	×	SE	24.06M	80.69	73.14
ResNet-50	✓	SE	24.07M	92.41	75.94
ResNet-50	×	NL	30.88M	74.14	74.77
ResNet-50	✓	NL	30.89M	91.72	77.21
ResNet-50	×	DP	26.12M	90.69	77.61
ResNet-50	✓	DP	26.13M	94.48	80.20

position for the two datasets, abbreviated as DPTAM in the following experiments.

3) *Different Fusion Strategies*: We also evaluate the experimental results of different fusion strategies, and the combination of components is shown as Fig. 6. We employ element-wise multiplication and average pooling on the channel axis instead of components in AFM to compose different fusion strategies. As shown in Table III, experiment 5, that is, MIE&RGB_AS_CAT, yields the best performance on both SLJD and Diving48 datasets. This is because MIE features mainly retain the information in the spatial dimension during the calculation process. In addition, if MIE and RGB features are fused by element-wise multiplication, their weak correlation in the channel dimension will bring noise to the fused features. Compared with other fusion strategies, AFM can effectively enhance the interaction between RGB features and MIE features.

4) *Comparison Against SE and NonLocal Block*: We notice that certain previous plug-and-play modules can improve the modeling ability of classification networks. One such module is squeeze-and-excitation (SE), which is initially introduced to extract global channel information in 2-D image tasks. To model spatiotemporal features, we extend it to its 3-D version. Another attention module is the NL, which is a typical self-attention implementation method used to obtain a global receptive field. Table IV compares the effects of several

TABLE V
EVALUATION OF THE DPTAM AND AFM

Exp	Model			SLJD		Diving48	
	Baseline	MIE&RGB	DPTAM	Top-1(%)	Top-5(%)	Top-1(%)	Top-5(%)
1	ResNet-50	Only RGB	×	65.17	92.75	73.75	97.05
2	ResNet-50	Only RGB	✓	90.69	98.99	77.61	96.54
3	ResNet-50	CAT	×	91.03	98.27	74.82	97.56
4	ResNet-50	CAT	✓	92.06	99.66	79.34	97.61
5	ResNet-50	AFM	×	94.13	99.31	75.53	97.21
6	ResNet-50	AFM	✓	94.48	99.66	80.20	98.17

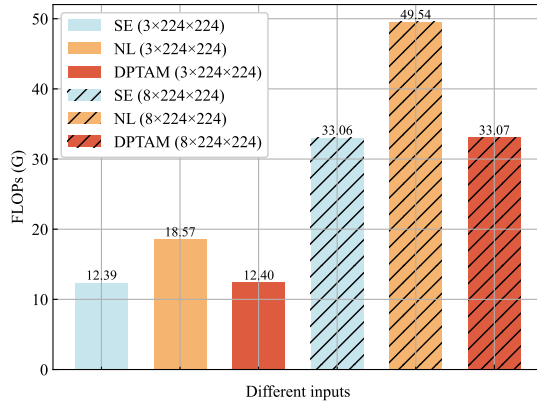


Fig. 7. Comparison of FLOPs with different inputs for different attention modules.

attention blocks on ResNet-50 and ResNet-50 with AFM. Our findings indicate that ResNet-50 with SE block results in lower accuracy on Diving48 than baseline. We believe this is because the SE block collects global information on the entire spatio-temporal axis, which destroys the fine-grained temporal structure needed for identifying subtle motion changes on the temporal axis. Although the NL block achieves higher accuracy than the SE block, it has 1.3 times more parameters, making it an inappropriate option. Our DP-Block has the highest accuracy among these attention blocks and is relatively inexpensive in terms of parameters. Furthermore, our experiments demonstrate that any module inserted into our baseline ResNet-50 with AFM performs better than on ResNet-50, indicating that AFM is effective without conflicts with other modules.

5) Comparison of FLOPs With Different Inputs for Different Attention Modules: We compare the computational cost (FLOPs) performance of different attention modules across various input frames. From Table IV and Fig. 7, we observe that DPTAM has more parameters than SE, but it effectively captures both local and global receptive fields, leading to improved accuracy without significantly increasing computation. In contrast, NL block or self-attention methods require more computation due to the embedding layer and matrix multiplication. Therefore, our DPTAM can be a simple and efficient component to extract spatial, temporal, and channel dependencies.

6) Effectiveness of the DPTAM and AFM: Here, the effectiveness of the DPTAM and AFM are verified separately, as far as Top-1 accuracy is concerned. As shown in Table V,

both modules contribute to the proposed MIE-Net. Either DPTAM or AFM are inserted, there is an improvement over the baseline. For example, experiment 4 achieves 1.0% and 4.5% higher accuracy compared with experiment 3 (92.0% versus 91.0% on SLJD, 79.3% versus 74.8% on Diving48), while experiment 6 achieved higher accuracy (94.4% versus 94.1% on SLJD, 80.2% versus 75.5% on Diving48) than experiment 5. Moreover, experiment 6 delivers absolute improvement of 2.4% and 0.9% over experiment 4 (94.4% versus 92.0% on SLJD, 80.2% versus 79.3% on Diving48). When combining the AFM and DPTAM, we achieve the best recognition accuracy on both SLJD and Diving48 datasets. The above results demonstrate that the recognition accuracy has been further improved by feature interaction and temporal modeling.

D. Comparison With State-of-the-Art

1) Results on SLJD: For this dataset, all experiments are trained on three NVIDIA GeForce GTX 1080 Ti GPUs using the PyTorch deep learning framework. Specifically, we have a comprehensive comparison with the state-of-the-art methods in terms of computational cost (FLOPs), the number of parameters (Params), Top-1 accuracy, and Top-5 accuracy, as shown in Table VI. First, compared with the initial 3-D CNN models, C3D [4], I3D [5], and I3D+NL [23] have lower Top-1 accuracy and higher FLOPs. 3-D kernels are obtained by appending temporal information to 2-D kernels, resulting in more computational cost, hence it is not an effective way to improve the network's capability. Later, TSN [26], TSM [7], and Slowfast [48] improve the modeling capability via different strategies, such as two streams or temporal modules. These methods generally achieve higher Top-1 and Top-5 accuracy with lower FLOPs than 3-D models. Notably, MIE-Net achieves better or more comparable performance than these methods. Furthermore, our method exceeds the temporal modeling capability of TAM [8] by 9.6% (90.6% versus 81.0%) on Top-1. Compared with the baseline method PAN [14], MIE-Net improves 14.4% higher accuracy (94.4% versus 80.0%) by temporal modeling and feature interaction. In addition, we also report several state-of-the-art methods, such as AIA [25], TCM [51], and STME [52]. Although these methods further consider the lightweight in model complexity (Params and FLOPs), the results demonstrate that MIE-Net achieves better performance. In a nutshell, MIE-Net not only effectively fuses multifeature information but also improves

TABLE VI
COMPARISON WITH THE STATE-OF-THE-ART ON SLJD

Method	Year	Pre-train	Frames	FLOPs	Params	Top-1(%)	Top-5(%)
C3D [4]	2015	Sports1M	16	66.3G	78.0M	81.03	98.28
I3D [5]	2017	ImageNet	32×2 clips	66.7G	27.2M	55.86	92.51
I3D [5]	2017	Kinetics400	32×2 clips	66.7G	27.2M	88.28	98.97
I3D+NL [23] ¹	2018	ImageNet	32×2 clips	75.0G	30.9M	57.93	86.21
TSN [26]	2018	ImageNet	8	33.0G	24.3M	73.79	95.17
TRN [47]	2018	ImageNet	8+48	33.0G	31.8M	85.17	99.31
TSM [7]	2019	ImageNet	8	33.0G	23.5M	80.34	98.62
TSM+NL [7] ¹	2019	ImageNet	8	49.5G	30.9M	84.83	98.28
Slowfast [48]	2019	Kinetics400	16×2 clips	27.9G	33.6M	80.69	99.31
TPN [49]	2020	ImageNet	8	50.5G	71.8M	83.79	98.96
TIN [10]	2020	ImageNet	8	34.0G	24.3M	76.55	99.31
PAN [14]	2021	ImageNet	3+3×4	13.5G	23.5M	80.00	98.62
TAM [8]	2021	ImageNet	3	13.4G	24.8M	81.03	98.27
TAM [8]	2021	ImageNet	8	33.1G	24.8M	81.96	95.08
TDN [50]	2021	ImageNet	8	36.1G	24.1M	86.20	99.55
AIA [25]	2022	ImageNet	3	12.5G	23.5M	82.41	98.96
AIA [25]	2022	ImageNet	8	33.2G	23.5M	81.72	99.31
TCM [51]	2022	ImageNet	3	12.6G	23.6M	86.11	98.61
TCM [51]	2022	ImageNet	8	33.5G	23.6M	81.94	95.83
STME [52]	2023	ImageNet	3	13.2G	24.2M	84.82	99.31
STME [52]	2023	ImageNet	8	35.2G	24.2M	86.20	99.65
DPTAM	Ours	ImageNet	3	12.4G	26.12M	90.69	98.99
MIE-Net	Ours	ImageNet	3+3×4	13.4G	26.13M	94.48	99.65

¹ NL denotes the NonLocal module.

TABLE VII
COMPARISON WITH THE STATE-OF-THE-ART ON SSV2

Method	Pre-train	Frames	FLOPs	Top-1(%)	Top-5(%)
TSN [26]	Kinetics400	8	23.9G	27.8	57.6
		16		30.0	60.5
TSN+TPN [49]	ImageNet	8	33G	55.2	-
TRN-M [47] ¹	ImageNet	8	33G	48.8	77.6
TRN-T [47] ¹	ImageNet	8+8	33G	55.5	83.1
Slowfast [48]	Kinetics400	4+32	67.5G	56.7	-
TSM [7]	Kinetics400	8	32.9G	58.7	84.8
FEXNet [24]	ImageNet	8	37G	60.1	-
TAM [8]	ImageNet	8	33G	60.5	86.2
TIN [10]	ImageNet	16	101G	60.0	85.5
AIA [25]	ImageNet	8	33.1G	60.3	86.4
PAN [14]	ImageNet	8+8×4	67.7G	60.8	86.7
TDN [50]	ImageNet	8	36G	64.0	88.8
TimeSformer [53]	ImageNet-1K	8×3	59G	59.5	-
TimeSformer-HR [53]	ImageNet-1K	16×3	511G	62.2	-
TimeSformer-L [53]	ImageNet-1K	96×3	714G	62.4	-
Swin-B [54]	Kinetics400	-	321G	69.6	92.7
MIE-Net	ImageNet	8+8×4	35.9G	60.9	86.9

¹ TRN-M denotes TRN Multiscale; TRN-T denotes TRN Two-stream.
“-” indicates the numbers are not available for us.

the modeling capability of the network, with little increase in computational cost (FLOPs) and the number of parameters (Params).

2) *Results on SSV2*: Table VII lists comparison results on SSV2, including MIE-Net, other 2-D CNN, 3-D CNN, and Transformer-based methods. Among them, TSN [26] and

TRN [47] use 2-D CNN and employ a late fusion strategy for long-range temporal modeling, but their results are not satisfactory. Compared with other 2-D CNN methods based on temporal attention such as TSM [7], TIN [10], and AIA [25], our method is superior to these methods, further demonstrating the advantages of MIE-Net in temporal modeling. The transformer-based methods achieve considerable performance, but the computational cost is dozens of times higher than our method. For example, Swin-B attains 69.6% Top-1 accuracy by using the Kinetics400 pre-trained model as initialization, but its computational cost is approximately ten times that of our method (321G versus 35.9G). Notably, some of these methods use high-end graphics cards with a batch size of 64 (e.g., TimeSformer uses Tesla V100 cards). By contrast, we can set eight frames with a batch size of eight due to equipment limitations. Nevertheless, our method achieves an overall appreciable performance by making a trade-off between accuracy and FLOPs.

3) *Results on Diving48*: To verify the effectiveness of our framework for action recognition in other transient sports, we compare the proposed MIE-Net with state-of-the-art methods on Diving48. As shown in Table VIII, MIE-Net achieves impressive performance on Diving48. Although TimeSformer [53] creates the highest scores on Top-1 when sampling 96 frames and performing three cropping (we only sample eight frames and perform center cropping), it is roughly 20 times the computational cost than ours. When we compare its results with eight frames, its Top-1 accuracy is lower than our method. We think the reason is that it focuses on long-term global modeling via a self-attention mechanism to overcome the shortcomings of the local receptive field

TABLE VIII
COMPARISON WITH THE STATE-OF-THE-ART ON DIVING48

Method	Pre-train	Frames	Top-1(%)
TimeSformer [53]	ImageNet-1K	8×3	74.9
TimeSformer-HR [53]	ImageNet-1K	16×3	78.0
TimeSformer-L [55]	ImageNet-1K	96×3	81.0
C3D [4]	ImageNet	16	34.5
P3D [6]	ImageNet	16	32.4
I3D [5]	Kinetics400	8	48.3
ST-S3D [19]	Kinetics400	8	50.6
Slowfast, R101 [48]	Kinetics400	16×3	77.6
TSN [26]	ImageNet	3	52.5
TSM [7]	ImageNet	8	76.9
GST [56]	ImageNet	16	38.8
EAN [29]	ImageNet	16×2	40.8
STC [57]	ImageNet	8	77.9
AIA [25]	ImageNet	8	79.4
TAM [8] ¹	ImageNet	8	76.1
PAN [14] ¹	ImageNet	8+8×4	78.1
TDN [50] ¹	ImageNet	8	78.7
MIE-Net	ImageNet	8+8×4	80.2

¹ denotes that it was done under the experimental settings of this paper.

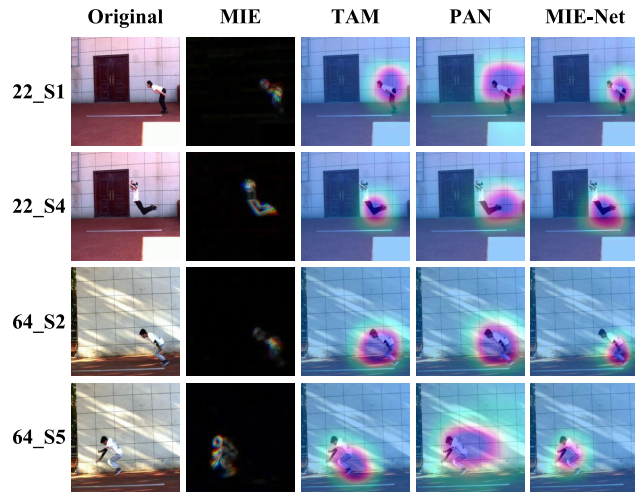


Fig. 8. Visualization of activation maps with CAM on SLJD. Images in classes (the first column) are verified with MIE (the second column), baseline TAM (the third column), baseline PAN (the fourth column), and MIE-Net (the last column).

of CNN. Therefore, it has an obvious advantage in long-term video action recognition by sampling many frames. In comparison, MIE-Net reaches a balance between accuracy and computational cost.

E. Visualization of Activation Maps

Class activation maps (CAMs) are used to identify the most important regions of an image that contribute to a specific classification decision made by a deep neural network. To confirm the effectiveness of our framework, we visualize the regions of interest of the network during feature extraction by utilizing the CAM [58] method. In this visualization, we train the networks with three sampled frames on SLJD and with eight sampled frames on Diving48. We only plot the

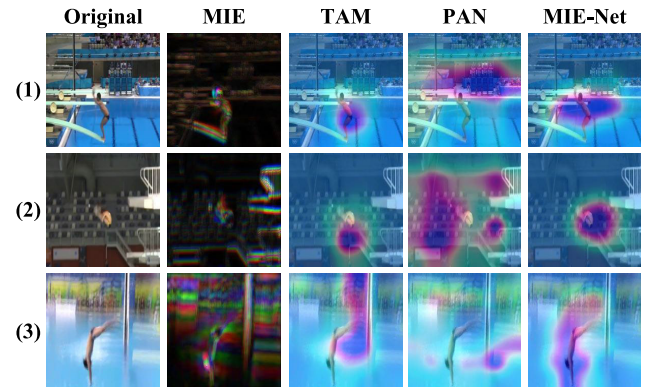


Fig. 9. Visualization of activation maps with CAM on Diving48. Images in classes (the first column) are verified with MIE (the second column), baseline TAM (the third column), baseline PAN (the fourth column), and MIE-Net (the last column).

activation maps in the central frame and present the saliency-enhanced maps. The CAMs of TAM [8] and PAN [14] and MIE-Net methods on SLJD are visualized in Fig. 8. The visualization results demonstrate that our method pays more attention to the motion boundaries and subtle information. As an illustration, the segment of 22_S1 is the process of swinging the arm downward to the waist, and the MIE image of 22_S1 depicts distinct movement changes occurring at the waist location. TAM and PAN images focus on larger regions, while MIE-Net produces more accurate results than other methods. Visualization analysis of the CAMs is conducted with some typical samples on Diving48, see Fig. 9. Examples (1), (2), and (3) are able to locate more action-related regions, thanks to our proposed MIE-Net with multi-feature fusion and temporal modeling.

VI. CONCLUSION

In this article, we proposed a video action recognition framework, MIE-Net, aiming at learning more fine-grained action information from RGB and motion features in an end-to-end manner. The AFM implements effective multi-feature interaction, while the DPTAM enhances the modeling capability during feature extraction. In addition, we collected some real-life SLJ videos that differ from existing datasets' videos in terms of strong background unbiasedness, to evaluate the effectiveness of our model robustly. Theoretical analysis and experimental results demonstrate that MIE-Net exhibits competitive performance. Although our study shows potential in video action recognition, a particular gap exists between theoretical advancements and practical application. In the future, we plan to study more lightweight models that are easy to deploy and extend them to more downstream video tasks, such as temporal action localization and spatio-temporal action detection, to make them more functional and practical.

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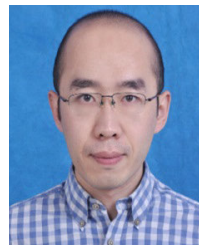
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